Python Basics

Sample knowledge-base document for RAG ingestion and retrieval testing.

1. Introduction to Python

Python is a high-level, general-purpose programming language designed with an emphasis on readability and developer productivity. It is widely used for automation, web development, data analysis, machine learning, artificial intelligence, testing, and scripting.

Key idea: Introduction to Python is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

2. Why Python Is Popular

Python has a simple syntax, a large standard library, and a very large ecosystem of third-party packages. Beginners can write useful programs quickly, while experienced developers can build large applications and data-processing systems.

Key idea: Why Python Is Popular is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

3. Variables and Data Types

A variable is a name that refers to a value. Common Python data types include integers, floating-point numbers, strings, booleans, lists, tuples, sets, dictionaries, and None. Python uses dynamic typing, so a variable does not need a type declaration before assignment.

Key idea: Variables and Data Types is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

4. Strings

Strings represent text. Python supports indexing, slicing, concatenation, searching, replacement, and formatting. Strings are immutable, which means operations that appear to modify a string actually create a new string.

Key idea: Strings is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

5. Lists and Collections

Lists are ordered and mutable collections. Tuples are ordered but immutable. Sets store unique elements and are useful for membership testing. Dictionaries store key-value pairs and are commonly used to represent structured information.

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6. Conditions and Loops

Conditional statements use if, elif, and else to make decisions. For loops iterate over sequences or other iterable objects. While loops repeat a block while a condition remains true. Break and continue can control loop execution.

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7. Functions

Functions package reusable logic into named blocks. They can accept parameters and return values. Functions improve readability, testing, reuse, and separation of responsibilities in a program.

Key idea: Functions is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

8. Modules and Packages

A module is a Python file containing reusable definitions. A package is a collection of modules. External packages can be installed with tools such as pip or uv. Examples include NumPy, Pandas, Requests, and Scikit-learn.

Key idea: Modules and Packages is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

9. Python for AI

Python is widely used in AI applications because of libraries for numerical computing, machine learning, deep learning, natural language processing, and LLM applications. A RAG system can use Python for document loading, chunking, embedding, retrieval, and generation.

Key idea: Python for AI is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.

10. Best Practices

Good Python code uses meaningful names, small functions, clear structure, exception handling, virtual environments, dependency management, and tests. Readability is important because software is maintained by people over time.

Key idea: Best Practices is an important foundational topic when learning and applying python basics. Understanding the concept, its purpose, and common examples helps when building practical software systems.